Deriving Feeding Rate from a DEB Model

1 Goal

This document explains how to derive a *feeding rate* f_{feed} (number of feeding motions per day) step-by-step from a Dynamic Energy Budget (DEB) model, given:

- A species-specific parameter set (κ_X, $\{\dot{p}_{Am}\}$)
- Environmental food availability (X, μ_X)
- Body size (L, V)

It is then the task of the behavioural control system to allocate these feeding motions within the day, subject to constraints such as bouts of repetitive feeding motions and pauses. In practice, feeding can be described as an oscillatory process with a (possibly time-varying) instantaneous frequency that integrates over 24 h to the total number f_{feed}.

Core parameters, state variables, and environmental quantities are listed in Appendix A.

2 Assumptions

We adopt two assumptions:

1. We focus on *rod-like isomorph larvae*, where uptake surfaces scale with L^2 under constant shape coefficients (Appendix B).
2. Each feeding motion ingests a *constant fraction* of body volume/area (to be decided which assumption is more realistic): $V_{\text{bite}} = \kappa_{\text{bite}} \cdot V$ or $V_{\text{bite}} = \kappa_{\text{bite}} \cdot L^2$.

3 Feeding frequency computation

Consumed food volume flux

Link between food and assimilation energy fluxes:

$$\dot{p}_A = \kappa_X \, \dot{p}_X \tag{1}$$

Food volume flux follows from food energy flux:

$$\dot{V}_X = \frac{\dot{p}_X}{\mu_X \, X} \tag{2}$$

Combining these:

$$\dot{V}_X = \frac{\dot{p}_A}{\kappa_X \, \mu_X \, X} \tag{3}$$

Unit check for Eq. (3):

$$[\dot{V}_X] = \frac{\text{energy/time}}{(\text{energy/C-mol})(\text{C-mol/volume})} = \text{volume/time}$$

Functional response and K

$$f(X) = \frac{X}{K + X}, \quad 0 \le f \le 1 \tag{4}$$

For an isomorph, assimilation scales with L^2:

$$\dot{p}_A = f\,\{\dot{p}_{Am}\}\,L^2 \tag{5}$$

Assimilation capacity relates to maximum ingestion as

$$\{\dot{p}_{Am}\} = \kappa_X\,\mu_X\,\{J_{Xm}\} \tag{6}$$

This yields

$$K = \frac{\{J_{Xm}\}}{\{\dot{F}_m\}} \tag{7}$$

Substitution of $\dot{p}_A$ into Eq. (3)

Start from Eq. (3) and substitute $\dot{p}_A$ using Eqs. (4)–(5):

$$\dot{V}_X = \frac{\dot{p}_A}{\kappa_X \mu_X X} = \frac{f\,\{\dot{p}_{Am}\}\,L^2}{\kappa_X \mu_X X} \tag{8}$$

Substitute $f(X)$ from Eq. (4) (the factor X cancels):

$$\dot{V}_X = \frac{\{\dot{p}_{Am}\}\,L^2}{\kappa_X \mu_X X}\,\frac{X}{K + X} = \frac{\{\dot{p}_{Am}\}\,L^2}{\kappa_X \mu_X\,(K + X)} \tag{9}$$

If desired, eliminate K using Eqs. (6)–(7). From Eq. (6),

$$\{J_{Xm}\} = \frac{\{\dot{p}_{Am}\}}{\kappa_X \mu_X} \tag{10}$$

so Eq. (7) gives

$$K = \frac{\{J_{Xm}\}}{\{\dot{F}_m\}} = \frac{\{\dot{p}_{Am}\}}{\kappa_X \mu_X\,\{\dot{F}_m\}} \tag{11}$$

Substituting Eq. (11) into Eq. (9) yields an explicit expression for the consumed food volume flux as a function of $\{\dot{p}_{Am}\}$ and X (with the remaining DEB constants shown):

$$\dot{V}_X = \frac{\{\dot{p}_{Am}\}\,L^2}{\kappa_X \mu_X \left(X + \frac{\{\dot{p}_{Am}\}}{\kappa_X \mu_X\,\{\dot{F}_m\}}\right)} = \frac{\{\dot{p}_{Am}\}\,L^2}{\kappa_X \mu_X X + \frac{\{\dot{p}_{Am}\}}{\{\dot{F}_m\}}} \tag{12}$$

Unit check for Eq. (12):

$$[\dot{V}_X] = \frac{[\{\dot{p}_{Am}\}]\,[L^2]}{[\kappa_X \mu_X X] + \left[\frac{\{\dot{p}_{Am}\}}{\{\dot{F}_m\}}\right]} = \frac{(\text{energy}\,\text{area}^{-1}\,\text{time}^{-1})(\text{area})}{\underbrace{(\text{energy C-mol}^{-1})(\text{C-mol volume}^{-1})}_{\kappa_X \mu_X X \sim \text{energy/volume}} + \underbrace{\frac{(\text{energy}\,\text{area}^{-1}\,\text{time}^{-1})}{(\text{volume}\,\text{area}^{-1}\,\text{time}^{-1})}}_{\{\dot{p}_{Am}\}/\{\dot{F}_m\} \sim \text{energy/volume}}} = \text{volume}\,\text{time}^{-1}$$

Number of feeding motions required

We define the feeding motion rate (feeding motions per unit time) as consumed food-volume flux divided by the volume per motion:

$$f_{\text{feed}} = \dot{N}_{bite} = \frac{\dot{V}_X}{V_{\text{bite}}} \tag{13}$$

Two alternative assumptions for V_{bite} are commonly considered in applications (to be decided empirically).

Alternative A (volume-fraction bite). Each feeding motion ingests a constant fraction of body volume V:

$$V_{\text{bite}} = \kappa_{\text{bite},V}\, V = \kappa_{\text{bite},V}\, L^3 \tag{14}$$

Combining Eqs. (13)–(14):

$$f_{\text{feed}} = \frac{\dot{V}_X}{\kappa_{\text{bite},V}\, V} = \frac{\dot{V}_X}{\kappa_{\text{bite},V}\, L^3} \tag{15}$$

Substituting $\dot{V}_X$ from Eq. (12) gives an explicit form in terms of $\{\dot{p}_{Am}\}$ and X:

$$f_{\text{feed}} = \frac{1}{\kappa_{\text{bite},V}\, L^3}\, \frac{\{\dot{p}_{Am}\}\, L^2}{\kappa_X \mu_X X + \frac{\{\dot{p}_{Am}\}}{\{\dot{F}_m\}}} = \frac{\{\dot{p}_{Am}\}}{\kappa_{\text{bite},V}\, L \left(\kappa_X \mu_X X + \frac{\{\dot{p}_{Am}\}}{\{\dot{F}_m\}}\right)} \tag{16}$$

Unit check for Eq. (16):

$$[f_{\text{feed}}] = \frac{[\dot{V}_X]}{[V_{\text{bite}}]} = \frac{\text{volume}\,\text{time}^{-1}}{[\kappa_{\text{bite},V}]\,[L^3]} = \frac{\text{volume}\,\text{time}^{-1}}{(1)\,(\text{volume})} = \text{time}^{-1}$$

Alternative B (area-proportional bite). Each feeding motion ingests a volume proportional to L^2:

$$V_{\text{bite}} = \kappa_{\text{bite},A}\, L^2 \tag{17}$$

Here $\kappa_{\text{bite},A}$ has units of length so that V_{bite} has units of volume.
Combining Eqs. (13) and (17):

$$f_{\text{feed}} = \frac{\dot{V}_X}{\kappa_{\text{bite},A}\, L^2} \tag{18}$$

Substituting $\dot{V}_X$ from Eq. (12) yields a size-cancelled form:

$$f_{\text{feed}} = \frac{1}{\kappa_{\text{bite},A}}\, \frac{\{\dot{p}_{Am}\}}{\kappa_X \mu_X X + \frac{\{\dot{p}_{Am}\}}{\{\dot{F}_m\}}} \tag{19}$$

Unit check for Eq. (19):

$$[f_{\text{feed}}] = \frac{[\dot{V}_X]}{[V_{\text{bite}}]} = \frac{\text{volume}\,\text{time}^{-1}}{[\kappa_{\text{bite},A}]\,[L^2]} = \frac{\text{volume}\,\text{time}^{-1}}{(\text{length})\,(\text{area})} = \frac{\text{volume}\,\text{time}^{-1}}{\text{volume}} = \text{time}^{-1}$$

A Parameters and variables

Species-specific DEB parameters

Notation	Description	Units	Equation
$\{\dot{p}_{Am}\}$	max assimilation rate per area	energy area^{-1} time^{-1}	$\{\dot{p}_{Am}\} = \kappa_X\, \mu_X\, \{\dot{J}_{Xm}\}$
$\{\dot{F}_m\}$	max searching/clearance rate per area	volume area^{-1} time^{-1}	–
$\{\dot{J}_{Xm}\}$	max ingestion rate per area	C-mol area^{-1} time^{-1}	–
κ_X	food-to-reserve conversion efficiency	–	$\dot{p}_A = \kappa_X\, \dot{p}_X$
$\kappa_{\text{bite},V}$	bite volume fraction (Alt. A)	–	$V_{\text{bite}} = \kappa_{\text{bite},V}\, V$
$\kappa_{\text{bite},A}$	bite area-proportional fraction (Alt. B)	length	$V_{\text{bite}} = \kappa_{\text{bite},A}\, L^2$
δ_M	DEB shape coefficient	–	$L = \delta_M L_w$

State variables

Notation	Description	Units	Equation
L	structural length	length	–
V	structural volume	volume	$V = L^3$

Environmental and derived quantities

Notation	Description	Units	Equation
X	food concentration	C-mol volume^{-1}	–
μ_X	energy per C-mol	energy C-mol^{-1}	–
K	half-saturation constant	C-mol volume^{-1}	$K = \{J_{Xm}\}/\{\dot{F}_m\}$
f	functional response	–	$f = X/(K + X)$
$\dot{p}_A$	assimilation flux	energy time^{-1}	$\dot{p}_A = f\, \{\dot{p}_{Am}\}\, L^2$
$\dot{p}_X$	food energy flux	energy time^{-1}	$\dot{p}_A = \kappa_X\, \dot{p}_X$
$\dot{V}_X$	food volume flux	volume time^{-1}	$\dot{V}_X = \dot{p}_A/(\kappa_X\, \mu_X X)$
V_{bite}	volume per feeding motion	volume	$V_{\text{bite}} = \kappa_{\text{bite},V}\, V$ or $\kappa_{\text{bite},A}\, L^2$
f_{feed}	feeding motions per time	time^{-1}	$f_{\text{feed}} = \dot{V}_X/V_{\text{bite}}$

B Larval body as a rod-like cylindrical isomorph

We treat the larval body as an *isomorph* in the DEB sense: shape is constant during growth, so structural *volume* scales with the cube of structural *length*, and structural *surface area* scales with the square of structural length.

For convenience (and because many larvae are approximately elongated), we represent the body as a *right circular cylinder* with a constant aspect ratio.

Notation note: we use L_w for observed (physical) length to avoid confusion with the DEB *scaled* length, which is commonly written as l.

Observed (physical) dimensions

Let L_w denote the observed body length (cm) and R the body radius (cm). We approximate structural volume V by the geometric cylinder volume

$$V = \pi R^2 L_w \tag{1}$$

The total cylinder surface area (including end caps) is

$$A_{\text{cyl}} = 2\pi R L_w + 2\pi R^2 \tag{2}$$

and for elongated larvae ($L_w \gg R$) the lateral area dominates:

$$A_{\text{cyl}} \approx 2\pi R L_w \tag{3}$$

Constant-shape (isomorph) constraint

We impose a constant aspect ratio

$$\alpha_{\text{AR}} \equiv \frac{L_w}{R} = \text{const} \tag{4}$$

so $R = L_w/\alpha_{\text{AR}}$ and therefore

$$V = \frac{\pi}{\alpha_{\text{AR}}^2} L_w^3 \tag{5}$$

Structural length and the DEB shape coefficient

In DEB, structural volume and structural length are related by

$$V = L^3 \tag{6}$$

For constant shape, observed length L_w is proportional to structural length L via the shape coefficient δ_M (dimensionless):

$$L = \delta_M \, L_w \tag{7}$$

For the cylindrical rod with constant aspect ratio α_{AR}, combining $V = L^3$ with $V = (\pi/\alpha_{\text{AR}}^2)L_w^3$ gives

$$\delta_M = \left(\frac{\pi}{\alpha_{\text{AR}}^2}\right)^{1/3} \tag{8}$$

Equivalently,

$$L_w = \frac{L}{\delta_M} = \left(\frac{\alpha_{\text{AR}}^2}{\pi}\right)^{1/3} L \tag{9}$$

Implications for DEB flux scaling

For an isomorph, DEB writes size dependence using structural length L with $V = L^3$ and (implicitly) structural surface area proportional to L^2. Below are the basic relations used in analyses.

Surface-area controlled fluxes and capacities ($\propto L^2$). Assimilation power (energy/time) is written with the surface-area-specific maximum $\{p_{Am}\}$:

$$\dot{p}_A = f\,\dot{p}_{Am} = f\,\{p_{Am}\}\,L^2 \tag{10}$$

Maximum assimilation power at size L is

$$\dot{p}_{Am} = \{p_{Am}\}\,L^2 \tag{11}$$

For feeding derived from searching/filtering (Chapter 2.1.4), the maximum searching rate scales as

$$\dot{F}_m = \{\dot{F}_m\}\,L^2 \tag{12}$$

and the maximum ingestion flux of food (amount/time) is

$$\dot{J}_{Xm} = \{\dot{J}_{Xm}\}\,L^2 \tag{13}$$

With food density X and half-saturation coefficient K, the (hyperbolic) functional response is

$$f = \frac{X}{K + X} \tag{14}$$

and ingestion follows

$$\dot{J}_X = f\,\dot{J}_{Xm} = f\,\{\dot{J}_{Xm}\}\,L^2 \tag{15}$$

The DEB textbook link between K, searching, and maximum ingestion is

$$K = \frac{\dot{J}_{Xm}}{\dot{F}_m} = \frac{\{\dot{J}_{Xm}\}}{\{\dot{F}_m\}} \tag{16}$$

Volume-controlled costs ($\propto L^3$). Somatic maintenance power is written using the volume-specific maintenance rate $[p_M]$:

$$\dot{p}_M = [p_M]L^3 \tag{17}$$

C Practical note – measuring κ_{bite}

In future experiments, the bite parameters can be estimated by combining measurements of body size, food volume flux $\dot{V}_X$, and feeding frequency f_{feed}.

For Alternative A (Eq. 14):

$$\kappa_{\text{bite},V} = \frac{\dot{V}_X}{f_{\text{feed}}\,V} \tag{18}$$

For Alternative B (Eq. 17):

$$\kappa_{\text{bite},A} = \frac{\dot{V}_X}{f_{\text{feed}}\,L^2} \tag{19}$$

At present we treat $\kappa_{\text{bite},V}$ or $\kappa_{\text{bite},A}$ as a species- and instar-specific quantity to be quantified in dedicated feeding trials.

References

- Kooijman, S.A.L.M. (2010). *Dynamic Energy Budget theory for metabolic organisation* (3rd ed.). Cambridge University Press.
- Standard DEB notation: `https://www.bio.vu.nl/thb/deb/deblab/add_my_pet/`